姓名：孔伟杰　学号：3190103782　HW2

(1)

(i) 系统辨识：　a_0、a_1、a_2、a_3、a_4

$$G_{\theta_r E}(s) = \frac{-\frac{1}{2} n_F K_T m_r L s}{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0}$$

$$a_0 = 9.8047$$

$$a_1 = 2.8522$$

$$a_2 = 0.5814$$

$$a_3 = 0.1093$$

$$a_4 = 0.0055$$

系统理论传递函数：

$$G_{\theta_r E}(s) == \frac{-0.3814s}{s^4 + 19.87s^3 + 105.7s^2 + 518.58s + 1782.7}$$

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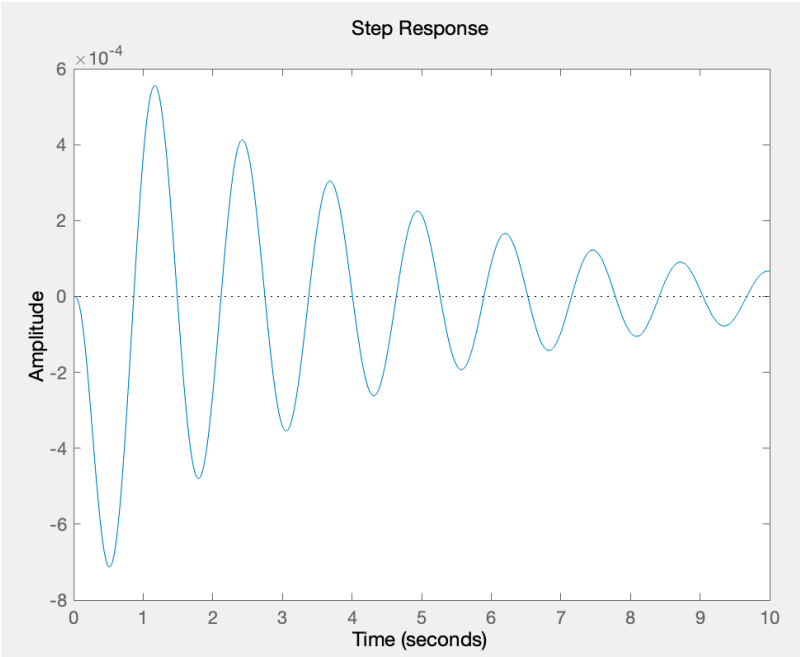
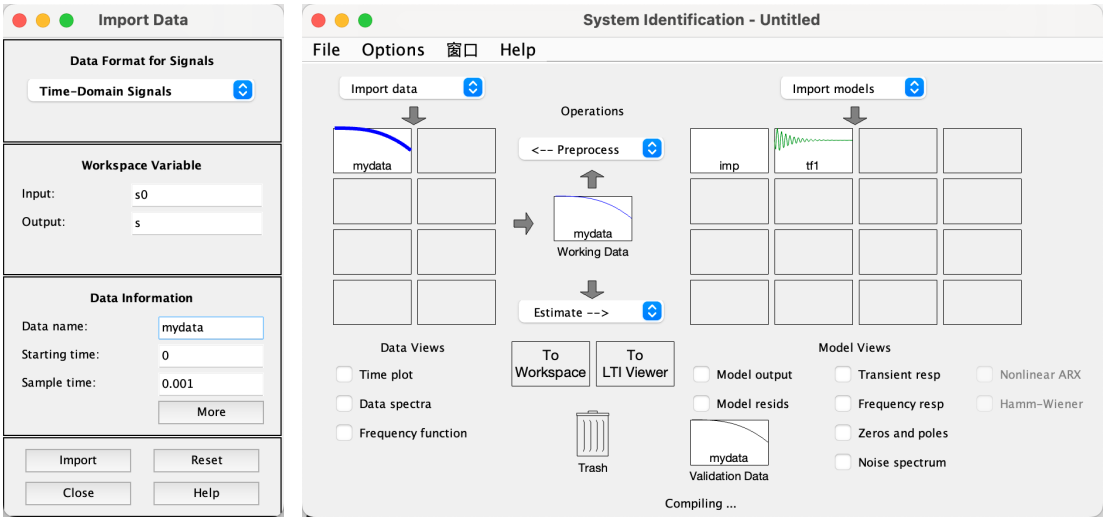
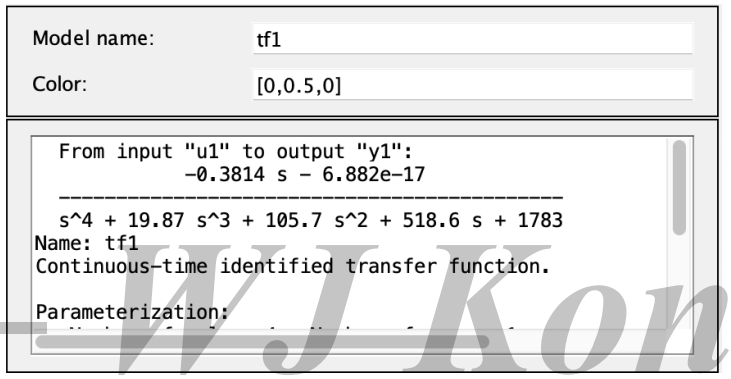
单位阶跃响应图：



取初始时间为 0s，采样周期为 0.001s，结束时间为 10s，输入为单位阶跃进行系统辨识：



系统辨识结果如下：



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实际辨识所得参数值与理论参数值数值十分接近，辨识有效。

（ii）系统辨识： c_0、c_1、d_0

$$G_{\theta_r X}(s) = \frac{d_0 s^2}{s^2 + c_1 s + c_0}$$

$$c_0 = \frac{\frac{1}{2} m_r g L}{I + \frac{1}{4} m_r L^2} = 22.99$$

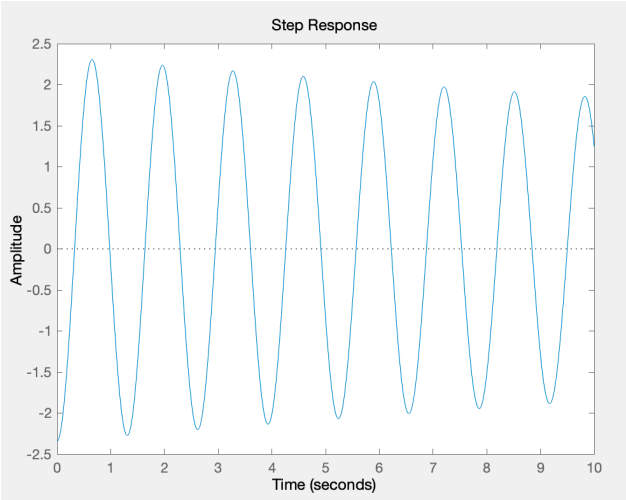
$$c_1 = \frac{b_r}{I + \frac{1}{4} m_r L^2} = 0.0476$$

$$d_0 = -\frac{\frac{1}{2} m_r L}{I + \frac{1}{4} m_r L^2} = -2.3437$$

系统理论传递函数：

$$G_{\theta_r X}(s) = \frac{-2.3437s^2}{s^2 + 0.0476s + 22.99}$$

单位阶跃响应图：



取初始时间为 0s，采样周期为 0.001s，结束时间为 10s，输入为单位阶跃进行系统辨识：

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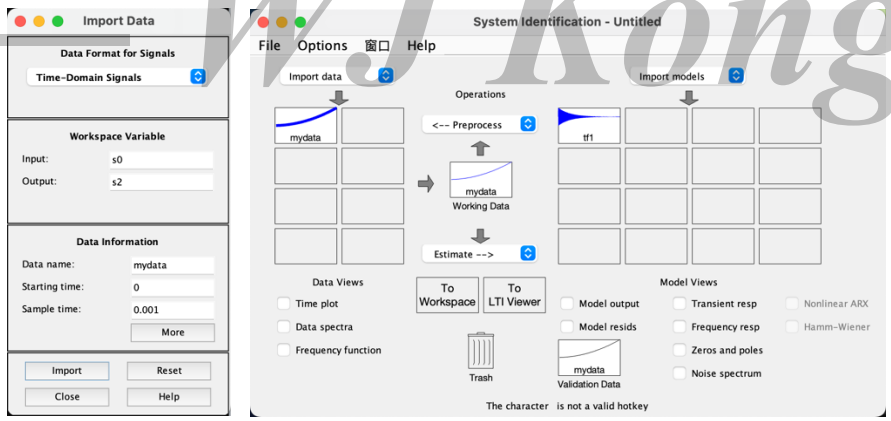
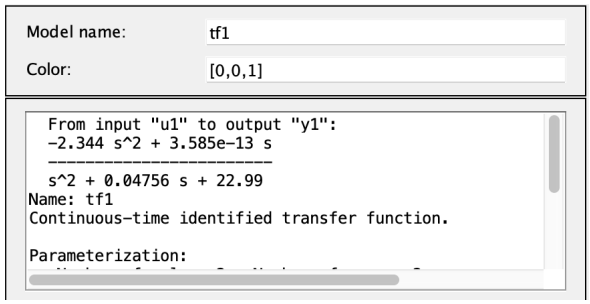


系统辨识结果如下：



实际辨识所得参数值与理论参数值数值十分接近，辨识有效。

(2)

$$G_{\theta_r F_c}(s) = \frac{\frac{1}{2} m_r L (L_A s + R_A) s}{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0}$$

$$a_0 = 9.8047$$

$$a_1 = 2.8522$$

$$a_2 = 0.5814$$

$$a_3 = 0.1093$$

$$a_4 = 0.0055$$

系统理论传递函数：

$$G_{\theta_r F_c}(s) = \frac{2.4182 s^2 + 34.9455 s}{s^4 + 19.87 s^3 + 105.7 s^2 + 518.1 s + 1781}$$

取初始时间为 0s，采样周期为 0.001s，结束时间为 10s

进行系统辨识：

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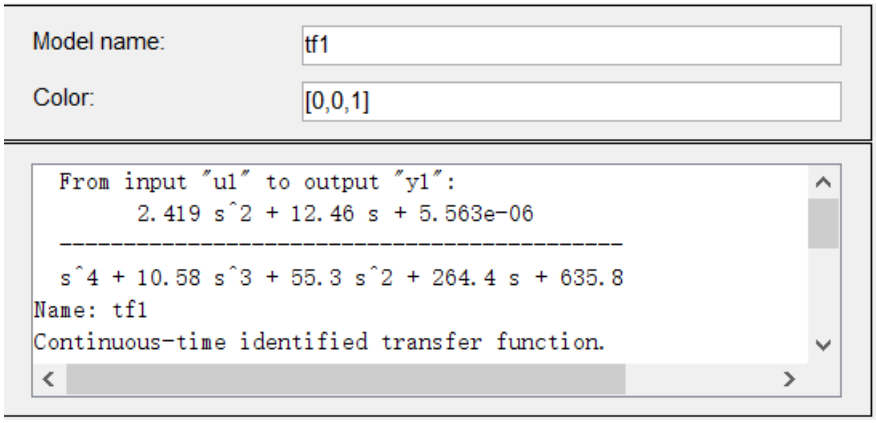
[1] 输入$F_c = 0.1N$时，系统辨识结果为：



[2] 输入$F_c = 1N$时，系统辨识结果为：

Model name: tf1

Color: [0,0,1]

```
  From input "u1" to output "y1":
        2.419 s^2 + 12.46 s + 5.563e-06
  ---------------------------------------------
  s^4 + 10.58 s^3 + 55.3 s^2 + 264.4 s + 635.8
Name: tf1
Continuous-time identified transfer function.
```

[3] 输入$F_c = 10N$时，系统辨识结果为：

Model name: tf1

Color: [0,0,1]

```
  From input "u1" to output "y1":
        2.415 s^2 + 12.46 s + 5.721e-06
  ---------------------------------------------
  s^4 + 10.58 s^3 + 55.34 s^2 + 264.6 s + 636.7
Name: tf1
Continuous-time identified transfer function.
```

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[4] 输入$F_c = 100N$时，系统辨识结果为：

Model name: tf1

Color: [0,0,1]

```
  From input "u1" to output "y1":
        0.02417 s^2 + 0.01233 s + 5.717e-10
  ------------------------------------------------
  s^4 + 1.052 s^3 + 0.5502 s^2 + 0.2629 s + 0.06295
Name: tf1
Continuous-time identified transfer function.
```

结论：摩擦力F_c越小，系统辨识后的结果越接近理论值